



GE Medical Systems

Technical Publications

Direction 2342628
Revision 0

Signa® System Milan MIL-S800i Switch Replacement

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

Direction 2128126 - Language Policy For Service Documentation, Rev 0

W A R N I N G

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは 英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
0.....	Oct 22, 2002	Initial version for Release.

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* This revision/letter corresponds to the indicated document's revision control system.

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1- MILAN MIL-S800I SWITCH REPLACEMENT

1-1 Tools Required

Medium Phillips screwdriver
Small flat head screwdriver
Container for screws

1-2 Cover Removal Procedure

Remove the Front Cover, the SGI Host Side Cover, of the OW Cabinet as shown.

1. Grasp the front cover from the sides (preferred) or from the top and bottom.
2. Pull the cover straight toward you. The ball and socket clips should release with as little as 8lbs. of pressure.
3. Always remove the cover over the SCSI box first. Loosen the rear thumbscrew but do not remove it.
4. Remove the Phillips head screw at the top of the front cover.



FRONT COVER SCREW LOCATION

Illustration 1-1

5. Lift the bottom of the cover slightly and slide the cover to the rear of the cabinet about 1/2" (13mm). Pull the cover toward you until the tabs clear the two top slots.



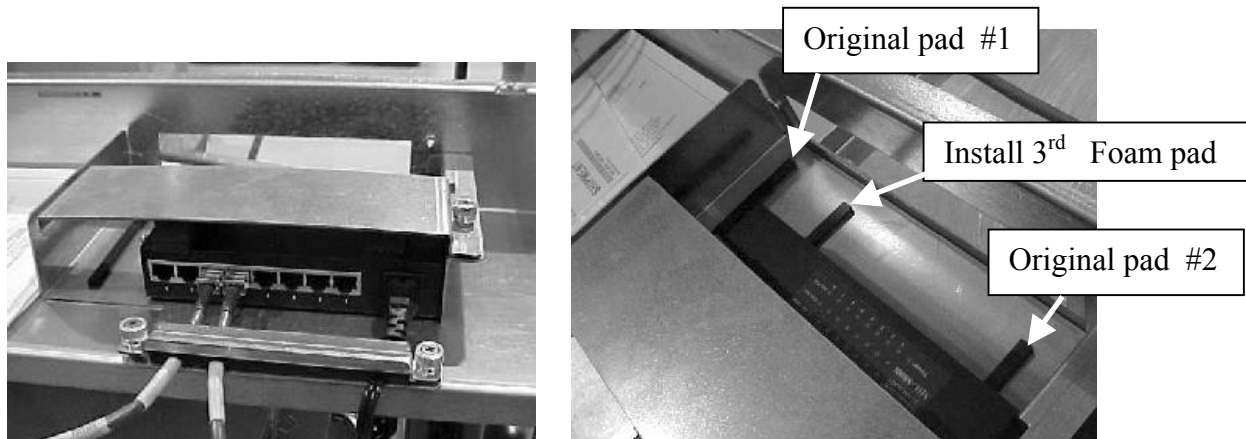
REMOVING SIDE COVER

ILLUSTRATION 1-2

6. Locate the current Hub/Switch on the left side of the Operator Workspace cabinet, above the PC.
7. On the front of the Hub, remove the two screws located on either end of the cable clamp. Set the cable clamp aside.
8. Disconnect the external cables and power cord.
9. Lift the old Hub/Switch out of the bracket shelf and set aside.

1-3 Installation Procedures

1. Remove the replacement Milan Switch and the foam pad from the original carton.
2. Install the foam pad so the Milan Switch is sitting on one of the old pads and the new one installed. Refer to illustration 1-3.



MILAN SWITCH AND FOAM PAD INSTALLATION
ILLUSTRATION 1-3

3. Install the switch. It should rest on one of the old pads and the new pad.
4. Reconnect the external cables and power cord to the unit.
5. Position the cables over the lower strain relief bracket of the cable clamp. Reposition the top bracket over the cables. Then install the two screws on either end.
6. Re-install the side cover and front cover.
7. Return power to the unit.
8. Place the old Hub or switch in the ESD Storage Bag and/or original carton, as applicable

2- TROUBLESHOOTING

2-1 Network Related Problems

1. Refer to Table 2-1 for LED functions.

Note

The Milan switch can be “ganged” together by connecting a RG-45 Cat 5 cables to any of the 8 ports. “Auto-negotiation” automatically sets the best possible bandwidth when connection is established with another network device. Each port supports Half/Full Duplex function.

TABLE 2-1
LED INDICATORS

LED Indicators	Color	Description
Power	Green	Lit: Power On Unlit: Power off
10/100 M Speed	Green	Lit: Link at 100 Mbs Unlit: Link at 10Mbs or no link
Link/Activity	Green	Lit: Good link on port. Blinking: Port receiving and transmitting Unlit: No link is occurring on the port
Full Duplex/Collison	Green	Lit: Full-Duplex mode Unlit: Half-duplex mode Blinking: Data collision

Loss of Communication with Magnet Monitoring

The Milan Switch does not have an MDI-X port. It uses straight through Network cables.

Note:

Some HFO and TwinSpeed systems previously used a crossover cable. There are three possible solutions:

1. Replace the crossover cable with a straight cable when using the Milan switch.
2. Add an adapter to change the crossed cable to a straight through cable.
3. Re-terminate the cable with a new connector on ONE end to make the existing cable a "straight through" 568B type configuration. (refer to Illustration 2-1).

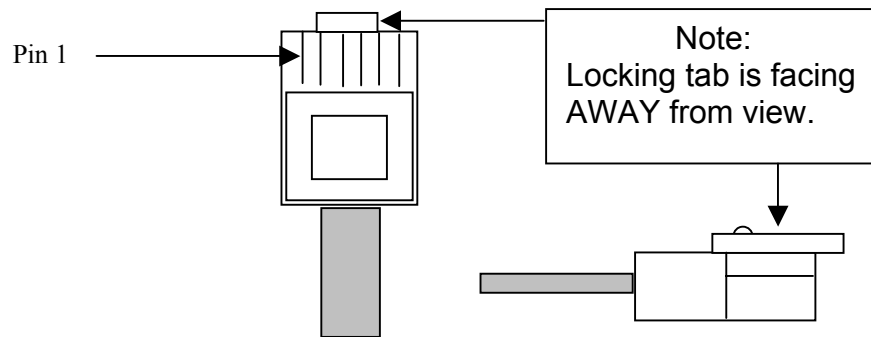
You can't use crossover cables with the Milan switch (turn it over and see the red circle w/ diag line over the symbol "X"). Using a crossover cable with the Milan switch will result in loss of connectivity. Just use straight cables with the Milan and it will auto sense the need to "internally" crossover.

All hub/switch ports are **MDI-X** meaning: Media Dependant Interface with internal crossover. This allows you to connect NICs to the hub/switch using a straight cable.

Most, but NOT all, hubs/switches have a special single port, which can be either MDI-X or MDI, but not both at the same time. When using this special port, you plug a **crossover cable into the connector labeled MDI** or plug a **straight cable into the connector labeled MDI-X**. There may be a physical switch on some Hubs/Switches that change the function of this special port. This special port is generally the FIRST or LAST port on the device.

This special port has direct access to the internal backbone of the hub/switch, so it's fast and is usually reserved/used for uplink only (i.e., connecting to another hub or switch).

Straight Through cable wiring		Crossover Cable Wiring	
RJ-45 connector pins:		RJ-45 connector pins:	
1 Tx+ ORN/WHT	1 Rc+ ORN/WHT	1 Rx+ ORANGE/WHITE	3 Tx+ GREEN/WHITE
2 Tx- ORN	2 Rc ORN	2 Rc- ORANGE	6 Tx- BROWN
3 Rc+ GRN/WHT	3 Tx+ GRN/WHT	3 Tx+ GREEN/WHITE	1 Rc+ ORANGE/WHITE
6 Rc- BRN	6 Tx- BRN	6 Tx- BROWN	2 Rc- ORANGE



568B WIRING CHART
ILLUSTRATION 2-1